

Linux Discretionary Access Control (DAC) & File Permissions Audit Report

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Environment / Tooling: Linux Bash Shell, POSIX File System Utilities

Target Path: /home/researcher2/projects (Research & Development Data Directory)

Security Standard: Principle of Least Privilege (PoLP), Discretionary Access Control (DAC) Hardening

Table of Contents

1.0 Project Overview & Security Objective	3
2.0 Directory Enumeration & Baseline Inspection	3
2.2 Baseline File System State (Pre-Remediation)	3
3.0 POSIX 10-Character Permission Notation Analysis	4
3.1 Analysis of current files permission	4
3.2 Audited Policy Violations and Threat Modeling:.....	4
4.0 Remediation & Hardening Implementation	6
4.1 Eliminating Global Public Write Access	6
4.2 Enforcing Strict Directory Isolation	6
4.3 Verification Matrix	7
5.0 Enterprise Hardening Best Practices	7

1.0 Project Overview & Security Objective

In an enterprise multi-tenant Linux environment, unmonitored file creations and lax default umasks frequently introduce **permission creep**, a state where unauthorized system entities gain read, write, or traversal rights over sensitive files. In research and intellectual property repositories, excessive file permission expose production data to tampering, unauthorized lateral reconnaissance, and data leakage.

As the assigned security analyst, an access control security audit of the */projects* **directory** was conducted to:

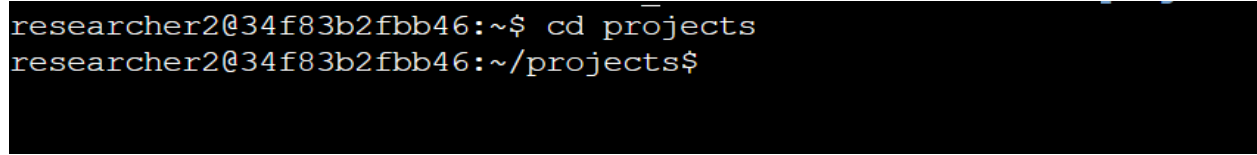
1. Enumerate standard and hidden filesystem assets across the research directory tree.
2. Identify and flag Discretionary Access Control (DAC) policy violations.
3. Apply targeted remediation using POSIX access-mode manipulation (chmod) to enforce strict isolation adhering to the **Principle of Least Privilege (PoLP)**.

2.0 Directory Enumeration & Baseline Inspection

2.1 Enumeration Execution

To evaluate current access baselines, directory hierarchy, and hidden system artifacts, the target directory was inspected using native Bash shell commands:

```
cd projects  
ls -la:
```

A terminal window with a black background and white text. The prompt is 'researcher2@34f83b2fbb46:~\$'. The first command entered is 'cd projects', and the second is 'ls -la'. The output of 'ls -la' is not visible in the screenshot.

```
researcher2@34f83b2fbb46:~$ cd projects  
researcher2@34f83b2fbb46:~/projects$
```

Figure 1.1 (Navigated to the targeted /Projects directory)

- ***cd projects***: Shifts the current working directory to the target project workspace.

2.2 Baseline File System State (Pre-Remediation)

The initial audit revealed the following permissions across the */projects* directory tree when the **ls -la** command was used:

- **ls -la**: . **ls** lists the files and subdirectory. **-l** formats output as an 8-column long listing revealing file types, access masks, link counts, ownership identifiers, byte

sizes, and timestamps. **-a** disables the suppression of dot-prefixed (.) hidden files and directories, ensuring complete visibility into concealed configuration files and metadata.

```
researcher2@42324f76486c:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 13:06 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 13:36 ..
-rw--w---- 1 researcher2 research_team  46 Jun 18 13:06 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 18 13:06 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 18 13:06 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 13:06 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 13:06 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 13:06 project_t.txt
researcher2@42324f76486c:~/projects$
```

Figure 2.0 (Existing files and access permissions)

3.0 POSIX 10-Character Permission Notation Analysis

3.1 Analysis of current files permission

To evaluate authorization states of the files and subdirectories in the **/project** directory to know who has access to what, the Linux 10-character notation string (**drwx--x---** and **-rw-rw-r--**) from **Figure 2.0 (/drafts** directory and **project_r.txt** file) was evaluated from left to right across four specific blocks:

- **d** is the first character indicating that the asset is a directory (d). **-** indicates that it is regular file.
- **rwX** are the second characters representing the user permission class. **r** indicates read, **w** indicates write, and **x** indicates execute privileges.
- **--x** are the third characters representing the group permission class. **--** indicate that the group has no read and write privileges while **x** indicates that the group has execute permission.
- **---** are the fourth characters that represent the other permission class. **---** from the **project_r.txt** file permission indicates that the **other** has no read, write and execute permission.

3.2 Audited Policy Violations and Threat Modeling:

During the file permission audit, the following policy violations were discovered in some files:

Asset Name	Mode String	Core Security Violation	Threat Vector & Business Impact
------------	-------------	-------------------------	---------------------------------

<i>project_k.txt</i>	-rw-rw-rw-	Global Public Write (o+w)	Permits any local user or compromised low-privilege service account to overwrite, tamper with, or manipulate project data.
.project_x.txt	-rw--w----	Asymmetric Group Write (g=w, g-r)	Group members possess blind write capabilities without <i>read</i> access, creating data-corruption risks and policy inconsistency.
drafts/	drwx--x---)	Group Traversal Exposure (g+x)	Group execution on a directory allows non-owner traversal and execution of files within if names are known, violating complete folder isolation.

```
researcher2@ef4f4a2ef133:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 12:17 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 12:31 ..
-rw--w---- 1 researcher2 research_team  46 Jun 18 12:17 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 18 12:17 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 18 12:17 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 12:17 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:17 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:17 project_t.txt
researcher2@ef4f4a2ef133:~/projects$
```

Figure 3.2.1(current project_k.txt file permission)

```
researcher2@ef4f4a2ef133:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 12:17 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 12:31 ..
-rw--w---- 1 researcher2 research_team  46 Jun 18 12:17 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 18 12:17 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 18 12:17 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 12:17 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:17 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:17 project_t.txt
researcher2@ef4f4a2ef133:~/projects$
```

Figure 3.2.2(current .project_x.txt file permission)

```
researcher2@34f83b2fbb46:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 07:22 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 08:00 ..
-rw--w---- 1 researcher2 research_team  46 Jun 18 07:22 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 18 07:22 drafts
-rw-rw-rw- 1 researcher2 research_team  46 Jun 18 07:22 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 07:22 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 07:22 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 07:22 project_t.txt
```

Figure 3.2.1(current drafts directory permission)

4.0 Remediation & Hardening Implementation

To remediate these exposures, symbolic mode manipulations via `chmod` (Change Mode) were executed against the affected assets.

4.1 Eliminating Global Public Write Access

Corporate security frameworks explicitly mandate that unauthenticated system entities (other) must never possess write or modification capabilities on active directory data files. To strip global write exposure from **project_k.txt**, the following command was executed:

```
researcher2@ef4f4a2ef133:~/projects$ chmod o-w project_k.txt | ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 12:17 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 12:31 ..
-rw--w---- 1 researcher2 research_team  46 Jun 18 12:17 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Jun 18 12:17 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:17 project_k.txt
-rw-r----- 1 researcher2 research_team  46 Jun 18 12:17 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:17 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 12:17 project_t.txt
researcher2@ef4f4a2ef133:~/projects$
```

The **chmod o-w** command targeted the **others** class in the **project_k.txt** file, explicitly terminating the **write** permission. The **pipe (/)** command was used with the **ls -la** to verify that the permission mask successfully transitioned from **-rw-rw-rw-** to the compliant state of **-rw-rw-r--**.

4.2 Enforcing Strict Directory Isolation

To ensure sensitive research drafts remain confidential, access to **/drafts** was restricted exclusively to the primary asset **owner (researcher2)**, revoking directory traversal from all group identities:

```
researcher2@34f83b2fbb46:~/projects$ chmod g-x drafts | ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 07:22 .
drwxr-xr-x 3 researcher2 research_team 4096 Jun 18 08:00 ..
-r--r----- 1 researcher2 research_team  46 Jun 18 07:22 .project_x.txt
drwx----- 2 researcher2 research_team 4096 Jun 18 07:22 drafts
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 07:22 project_k.txt
-rw----- 1 researcher2 research_team  46 Jun 18 07:22 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 07:22 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 Jun 18 07:22 project_t.txt
researcher2@34f83b2fbb46:~/projects$
```

Figure 4.2 (restricted access to /drafts to the primary asset owner (researcher2))

- **chmod g-x:** Modifies the *group* (g) category by removing (-) the execute/traversal (x) bit.
- **ls -la:** Evaluates directory permissions directly without recursing into child inodes. Permissions transitioned from **drwx--x---** to **drwx-----**. Only **researcher2** retains directory indexing, **read**, **write**, and traversal rights; group and other entities are denied entry.

4.3 Verification Matrix

Target Asset	Applied Mode	Security Objective Achieved
project_k.txt	chmod o-w	Terminated unauthorized external modification; integrity protected.
drafts/	chmod g-x	Enforced directory isolation; prevented group reconnaissance.
.project_x.txt	chmod u-w, g=r	Eliminated inconsistent blind-write flags; preserved historical reference state.

5.0 Enterprise Hardening Best Practices

To prevent similar permission drift and maintain secure-by-default access across multi-user environments:

1. **Default Umask Configuration:** Configure corporate Linux login profiles with a secure umask to eliminate automatic global read/write exposures.
2. **Automated Audit Cron Jobs:** Deploy automated file integrity and permissions audits to identify world-writable files across sensitive directories.

The enforcement of these discretionary access controls measures minimize the internal threat landscape, ensuring comprehensive protection of corporate data assets from unauthorized modification or data leakage